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Group:SE2

LAB_3

PART A1: Containers, Namespaces & Cgroups

```
PS C:\WINDOWS\system32> docker run --rm -it --name nsdemo ubuntu:22.04 bash

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/nsdemo" is already in use by container "fb97af2f8f62e902467c9
5eb90f99d6d36f4dc64ef2f8324564742ffa684e879". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
PS C:\WINDOWS\system32> docker rm -f nsdemo
nsdemo
PS C:\WINDOWS\system32> docker run --rm -it --name nsdemo ubuntu:22.04 bash
root@0cab6a0e75f4:/# hostname
0cab6a0e75f4
root@0cab6a0e75f4:/# ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 12:11 pts/0    00:00:00 bash
root          10       1  0 12:12 pts/0    00:00:00 ps -ef
```

```
root@0cab6a0e75f4:/# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 42:8b:9a:15:4e:d0 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.2/16 brd 172.17.255.255 scope global eth0
       valid_lft forever preferred_lft forever
root@0cab6a0e75f4:/# mount | head
overlay on / type overlay (rw,relatime,lowerdir=/var/lib/desktop-containerd/daemon/io.containerd.snapshotter.v1.overlayfs/snapsh
ots/335/fs:/var/lib/desktop-containerd/daemon/io.containerd.snapshotter.v1.overlayfs/snapshots/331/fs,upperdir=/var/lib/desktop-
containerd/daemon/io.containerd.snapshotter.v1.overlayfs/snapshots/336/fs,workdir=/var/lib/desktop-containerd/daemon/io.containe
rd.snapshotter.v1.overlayfs/snapshots/336/work)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
tmpfs on /sys/fs/cgroup type tmpfs (rw,nosuid,nodev,noexec,relatime,mode=755)
cgroup on /sys/fs/cgroup/cpuset type cgroup (ro,nosuid,nodev,noexec,relatime,cpuset)
cgroup on /sys/fs/cgroup/cpu type cgroup (ro,nosuid,nodev,noexec,relatime,cpu)
cgroup on /sys/fs/cgroup/cpuacct type cgroup (ro,nosuid,nodev,noexec,relatime,cpuacct)
cgroup on /sys/fs/cgroup/blkio type cgroup (ro,nosuid,nodev,noexec,relatime,blkio)
root@0cab6a0e75f4:/# cat /proc/1/cgroup
15:name=systemd:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
14:misc:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
13:rdma:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
12:pids:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
11:hugetlb:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
```

```
root@0cab6a0e75f4:/# cat /proc/1/cgroup
15:name=systemd:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
14:misc:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
13:rdma:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
12:pids:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
11:hugetlb:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
10:net_prio:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
9:perf_event:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
8:net_cls:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
7:freezer:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
6:devices:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
5:memory:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
4:blkio:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
3:cpuacct:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
2:cpu:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
1:cpuset:/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
0::/docker/0cab6a0e75f47786c3b8c0abb0f0fd41cc64acad7317b695c4954de389d2f564
root@0cab6a0e75f4:/#
```

A2 – Host vs container process

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>docker ps
CONTAINER ID   IMAGE          COMMAND   CREATED          STATUS          PORTS     NAMES
1d78cf5a28dd   ubuntu:22.04   "bash"    18 seconds ago   Up 17 seconds             nsdemo

C:\Users\hp>docker inspect nsdemo --format "{{.State.Pid}}"
1813

C:\Users\hp>ps -fp 1813
'ps' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\hp>docker top nsdemo
UID                 PID                 PPID                C                   STIME               TTY                 TIME
       CMD
root                1813                1787                0                   12:32               pts/0               00:00:00
       bash

C:\Users\hp>
```

A3 – Resource limits (cgroups)

```
root@5d715ae9c1fa: /

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>docker run --rm -it --cpus="0.5" --memory="256m" ubuntu:22.04 bash
root@5d715ae9c1fa:/# apt-get update
Get:1 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1292 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [61.6 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6862 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1601 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [69.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7066 kB]
Get:15 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3863 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4191 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [82.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.7 kB]
Fetched 45.5 MB in 14s (3220 kB/s)
Reading package lists... Done
root@5d715ae9c1fa:/# apt-get install -y stress
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Fetched 18.4 kB in 0s (68.4 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package stress.
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
Setting up stress (1.0.5-1) ...
root@5d715ae9c1fa:/# stress --cpu 2 --vm 1 --vm-bytes 220M --timeout 30
stress: info: [235] dispatching hogs: 2 cpu, 0 io, 1 vm, 0 hdd
stress: info: [235] successful run completed in 31s
root@5d715ae9c1fa:/#
```

```
Command Prompt - docker s

CONTAINER ID   NAME             CPU %     MEM USAGE / LIMIT   MEM %     NET I/O           BLOCK I/O   PIDS
5d715ae9c1fa   boring_almeida   0.00%     1.215MiB / 256MiB   0.47%     51.3MB / 1.23MB   0B / 0B     1
1d78cf5a28dd   nsdemo           0.00%     872KiB / 7.445GiB   0.01%     1.34kB / 126B     0B / 0B     1
```

PART B :Docker Image Layering

```python
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Lab 3"

@app.route("/health")
def health():
    return "ok"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
flask==3.0.0
```

B2:Basic Docker image

```
Debug this build failure with Gordon → docker ai "help me fix this build failure"

C:\Users\hp\lab3>notepad requirements.txt

C:\Users\hp\lab3>notepad app.py

C:\Users\hp\lab3>docker build -f Dockerfile.basic -t lab3-basic .
[+] Building 40.0s (10/10) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile.basic                                                0.0s
 => => transferring dockerfile: 196B                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                      14.9s
 => [internal] load .dockerignore                                                                         0.0s
 => => transferring context: 2B                                                                           0.0s
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e45  16.9s
 => => resolve docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e459  0.0s
 => => sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357 248B / 248B                 0.3s
 => => sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65 12.10MB / 12.10MB           5.3s
 => => sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d 1.29MB / 1.29MB             2.5s
 => => sha256:3531af2bc2a9c888375465278 3cf96207d53189db279c9637b7157d034de7ecd 29.78MB / 29.78MB          12.2s
 => => extracting sha256:3531af2bc2a9c888375465278 3cf96207d53189db279c9637b7157d034de7ecd                2.8s
 => => extracting sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d                 0.2s
 => => extracting sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65                 1.4s
 => => extracting sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357                 0.0s
 => [internal] load build context                                                                         0.1s
 => => transferring context: 330B                                                                         0.0s
 => [2/5] WORKDIR /app                                                                                    0.8s
 => [3/5] COPY requirements.txt .                                                                         0.1s
 => [4/5] RUN pip install -r requirements.txt                                                             5.0s
 => [5/5] COPY app.py .                                                                                   0.1s
 => exporting to image                                                                                    1.7s
 => => exporting layers                                                                                   0.9s
 => => exporting manifest sha256:c5b6482906c17d383b31f4333c03abcfaacc5809a7e528929adba211fb0a2205          0.0s
 => => exporting config sha256:651ca5d25921b58f57d74a493076d3d8c4fc23f6e8ad7e24cb020589457bcbc7           0.0s
 => => exporting attestation manifest sha256:1df38d3f3180b75d06eb40e17f4c138fadbe9bde7ea5a7d30bdd1fb1604e1623  0.0s
 => => exporting manifest list sha256:72a0651790baaae00983d1acdb5720dd54e5d3f77a24a60804e1e757fb22a328    0.0s
 => => naming to docker.io/library/lab3-basic:latest                                                      0.0s
 => => unpacking to docker.io/library/lab3-basic:latest                                                   0.7s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/0iatsz5gbqgke39j35rsxwt5v

C:\Users\hp\lab3>docker history lab3-basic
IMAGE          CREATED              CREATED BY                                      SIZE      COMMENT
72a0651790ba   About a minute ago   CMD ["python" "app.py"]                         0B        buildkit.dockerfile.v0
<missing>      About a minute ago   EXPOSE [5000/tcp]                               0B        buildkit.dockerfile.v0
<missing>      About a minute ago   COPY app.py . # buildkit                        12.3kB    buildkit.dockerfile.v0
<missing>      About a minute ago   RUN /bin/sh -c pip install -r requirements.t…   16.5MB    buildkit.dockerfile.v0
<missing>      About a minute ago   COPY requirements.txt . # buildkit              12.3kB    buildkit.dockerfile.v0
<missing>      About a minute ago   WORKDIR /app                                    8.19kB    buildkit.dockerfile.v0
<missing>      3 days ago           CMD ["python3"]                                 0B        buildkit.dockerfile.v0
<missing>      3 days ago           RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB    buildkit.dockerfile.v0
<missing>      3 days ago           RUN /bin/sh -c set -eux;   savedAptMark="${a…   41.3MB    buildkit.dockerfile.v0
<missing>      3 days ago           ENV PYTHON_SHA256=c08bc65a81971c1dd578318282…   0B        buildkit.dockerfile.v0
<missing>      3 days ago           ENV PYTHON_VERSION=3.12.13                      0B        buildkit.dockerfile.v0
<missing>      3 days ago           ENV GPG_KEY=7169605F62C751356D054A26A821E680…   0B        buildkit.dockerfile.v0
<missing>      3 days ago           RUN /bin/sh -c set -eux;  apt-get update;  a…   4.94MB    buildkit.dockerfile.v0
<missing>      3 days ago           ENV LANG=C.UTF-8                                0B        buildkit.dockerfile.v0
<missing>      3 days ago           ENV PATH=/usr/local/bin:/usr/local/sbin:/usr…   0B        buildkit.dockerfile.v0
<missing>      4 days ago           # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.4MB    debuerreotype 0.17

C:\Users\hp\lab3>docker image ls
                                                                                    Info →   U  In Use
IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
lab3-basic:latest  72a0651790ba     199MB         49MB
ubuntu:22.04       962f6cadeae0     119MB         31.7MB    U
ubuntu:latest      c4a8d5503dfb     117MB         29.7MB    U
```

B3:Multi-stage build

```
C:\Users\hp\lab3>notepad Dockerfile.multistage

C:\Users\hp\lab3>docker build -f Dockerfile.multistage -t lab3-multi .
[+] Building 7.9s (12/12) FINISHED                                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile.multistage                                                                                     0.0s
 => => transferring dockerfile: 312B                                                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                                                 1.0s
 => [internal] load .dockerignore                                                                                                                   0.0s
 => => transferring context: 2B                                                                                                                     0.0s
 => [internal] load build context                                                                                                                   0.0s
 => => transferring context: 63B                                                                                                                    0.0s
 => CACHED [builder 1/4] FROM docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e4590c18c2e3            0.0s
 => => resolve docker.io/library/python:3.12-slim@sha256:46cb7cc2877e60fbd5e21a9ae6115c30ace7a077b9f8772da879e4590c18c2e3                          0.0s
 => CACHED [stage-1 2/4] WORKDIR /app                                                                                                               0.0s
 => [builder 2/4] WORKDIR /build                                                                                                                    0.0s
 => [builder 3/4] COPY requirements.txt .                                                                                                           0.0s
 => [builder 4/4] RUN pip install --prefix=/install -r requirements.txt                                                                             6.1s
 => [stage-1 3/4] COPY --from=builder /install /usr/local                                                                                           0.1s
 => [stage-1 4/4] COPY app.py .                                                                                                                     0.1s
 => exporting to image                                                                                                                              0.8s
 => => exporting layers                                                                                                                             0.4s
 => => exporting manifest sha256:e609d8c8027b2b0744cb25bdf4b56bc9809ea398b761d3f987cf746831022a285d                                                 0.0s
 => => exporting config sha256:75f5135f833ed3a3a33ce07bf8b127236b4a8816c2fc0f7e8133588e7066aa066                                                    0.0s
 => => exporting attestation manifest sha256:0412588eb50caa976adb160782c719ba07d56e854bc1a858773ebc76bcae2c28                                      0.0s
 => => exporting manifest list sha256:64088a2a4ce3d23422fa9969015e00f84f0f3b0c1f7a9b2a91471e377362964cc                                            0.0s
 => => naming to docker.io/library/lab3-multi:latest                                                                                                0.0s
 => => unpacking to docker.io/library/lab3-multi:latest                                                                                             0.2s

 View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/wg8rtt7auon29phlclbnsv920

C:\Users\hp\lab3>docker image ls

                                                                                                                                                    i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
lab3-basic:latest    72a0661790ba       199MB         49MB
lab3-multi:latest    6488a2a4ce3d       184MB         44.7MB
ubuntu:22.04         962f6cadoaa0       119MB         31.7MB    U
ubuntu:latest        c4a8d5503dfb       117MB         29.7MB    U

C:\Users\hp\lab3>docker history lab3-multi
IMAGE          CREATED        CREATED BY                                      SIZE      COMMENT
6488a2a4ce3d   21 seconds ago   CMD ["python" "app.py"]                       0B        buildkit.dockerfile.v0
<missing>      21 seconds ago   EXPOSE [5000/tcp]                             0B        buildkit.dockerfile.v0
<missing>      21 seconds ago   COPY app.py . # buildkit                      12.3kB    buildkit.dockerfile.v0
<missing>      21 seconds ago   COPY /install /usr/local # buildkit           5.51MB    buildkit.dockerfile.v0
<missing>      4 minutes ago    WORKDIR /app                                  8.194kB   buildkit.dockerfile.v0
<missing>      3 days ago       CMD ["python3"]                               0B        buildkit.dockerfile.v0
<missing>      3 days ago       RUN /bin/sh -c set -eux;  for src in idle3 p… 16.4kB    buildkit.dockerfile.v0
<missing>      3 days ago       RUN /bin/sh -c set -eux;   savedAptMark="$(a… 41.3MB    buildkit.dockerfile.v0
<missing>      3 days ago       ENV PYTHON_SHA256=c08bc65a81971c1dd5783182827… 0B       buildkit.dockerfile.v0
<missing>      3 days ago       ENV PYTHON_VERSION=3.12.11                    0B        buildkit.dockerfile.v0
<missing>      3 days ago       ENV GPG_KEY=7169605F62C751356D054A26A821680… 0B        buildkit.dockerfile.v0
<missing>      3 days ago       RUN /bin/sh -c set -eux;  apt-get update;  a… 4.94MB    buildkit.dockerfile.v0
<missing>      3 days ago       ENV LANG=C.UTF-8                              0B        buildkit.dockerfile.v0
<missing>      3 days ago       ENV PATH=/usr/local/bin:/usr/local/sbin:/usr… 0B        buildkit.dockerfile.v0
<missing>      4 days ago       # debian.sh --arch 'amd64' out/ 'trixie' '@1… 87.4MB    debuerreotype= 0.17

C:\Users\hp\lab3>
```

PART C :

C1 – Cluster creation

```
Administrator: Windows PowerShell
runtime.goexit
        runtime/asm_amd64.s:1693
PS C:\WINDOWS\system32> kubectl cluster-info
Kubernetes control plane is running at https://kubernetes.docker.internal:6443
CoreDNS is running at https://kubernetes.docker.internal:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\WINDOWS\system32> kubectl get nodes -o wide
NAME             STATUS   ROLES           AGE     VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE         KERNEL-VERSION                    CONTAINER-RUNTIME
docker-desktop   Ready    control-plane   4h56m   v1.34.1   192.168.65.3   <none>        Docker Desktop   5.15.167.4-microsoft-standard-WSL2   docker://29.4.0
PS C:\WINDOWS\system32>
```

C2:

```
Administrator: Windows PowerShell
 => => exporting layers                                                                                                    0.5s
 => => exporting manifest sha256:f3341e3a1af5baa78d81b7dd6a674ac0bba168fa1160501bcc214a539031c976                          0.0s
 => => exporting config sha256:7a24cdf0042e4af50b3cca63b7f5b8a7dd15398ddd7eb16ac707458de0e44dc2                            0.0s
 => => exporting attestation manifest sha256:bf01553c3102aebd66e9e466b2cf489f1a676b7f734d62660baddf67b1527f19              0.0s
 => => exporting manifest list sha256:c3909f1614640da80471d9900cb33994ca5445be9a5c7e0552fd2b78161cf00d                     0.0s
 => => naming to docker.io/library/lab3-multi:latest                                                                       0.0s
 => => unpacking to docker.io/library/lab3-multi:latest                                                                    0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ioj9y23n5iq47ktm03v5y34mw
PS C:\Users\hp\lab3> notepad deployment.yaml
PS C:\Users\hp\lab3> notepad service.yaml
PS C:\Users\hp\lab3> dir


    Directory: C:\Users\hp\lab3


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        25/04/2026  03:47 PM            238 app.py
-a----        25/04/2026  09:52 PM            368 deployment.yaml
-a----        25/04/2026  03:46 PM            151 Dockerfile.basic
-a----        25/04/2026  03:52 PM            262 Dockerfile.multistage
-a----        25/04/2026  03:46 PM             12 requirements.txt
-a----        25/04/2026  09:53 PM            166 service.yaml


PS C:\Users\hp\lab3> kind load docker-image lab3-basic --name lab3-cluster
ERROR: no nodes found for cluster "lab3-cluster"
PS C:\Users\hp\lab3> kubectl apply -f deployment.yaml
deployment.apps/lab3-web created
PS C:\Users\hp\lab3> kubectl apply -f service.yaml
>>
service/lab3-web-svc created
PS C:\Users\hp\lab3> kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
lab3-web-868cd7fbff-7qv2l   1/1     Running   0          23s
lab3-web-868cd7fbff-j4tlp   1/1     Running   0          23s
lab3-web-868cd7fbff-wns8v   1/1     Running   0          23s
PS C:\Users\hp\lab3>
```

C3:

```
Administrator: Windows PowerShell
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                                                                                                   Install the latest PowerShell for new features and
mprovements! https://aka.ms/PSWindows
                PS C:\WINDOWS\system32> curl http://localhost:9090/
Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is
parsed.
      RECOMMENDED ACTION:
      Use the -UseBasicParsing switch to avoid script code execution.

      Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): A


StatusCode        : 200
StatusDescription : OK
Content           : Hello from Lab 3
RawContent        : HTTP/1.1 200 OK
                    Connection: close
                    Content-Length: 16
                    Content-Type: text/html; charset=utf-8
                    Date: Sat, 25 Apr 2026 18:59:09 GMT
                    Server: Werkzeug/3.1.8 Python/3.12.13

                    Hello from Lab 3
Forms             : {}
Headers           : {[Connection, close], [Content-Length, 16], [Content-Type, text/html; charset=utf-8], [Date, Sat,
                    25 Apr 2026 18:59:09 GMT]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 16


PS C:\WINDOWS\system32> Invoke-RestMethod http://localhost:9090/health
ok
PS C:\WINDOWS\system32>
```

C4 :Access service

```
PS C:\Users\hp\lab3> kubectl port-forward service/lab3-web-svc 9090:80
Forwarding from 127.0.0.1:9090 -> 5000
Forwarding from [::1]:9090 -> 5000
Handling connection for 9090
Handling connection for 9090
```

PART D:

D1:Node labelling:

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\WINDOWS\system32> kubectl label nodes --all node-role=general
node/docker-desktop labeled
PS C:\WINDOWS\system32> kubectl get nodes --show-labels
NAME             STATUS   ROLES           AGE     VERSION   LABELS
docker-desktop   Ready    control-plane   5h50m   v1.34.1   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes
.io/hostname=docker-desktop,kubernetes.io/os=linux,node-role.kubernetes.io/control-plane=,node-role=general,node.kubernetes.io/exclude-from-external-load
-balancers=
PS C:\WINDOWS\system32>
```

D2:

```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> kubectl label nodes --all node-role=general
node/docker-desktop labeled
PS C:\WINDOWS\system32> kubectl get nodes --show-labels
NAME             STATUS   ROLES           AGE     VERSION   LABELS
docker-desktop   Ready    control-plane   5h50m   v1.34.1   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes
.io/hostname=docker-desktop,kubernetes.io/os=linux,node-role.kubernetes.io/control-plane=,node-role=general,node.kubernetes.io/exclude-from-external-load
-balancers=
PS C:\WINDOWS\system32> cd C:\Users\hp\lab3
PS C:\Users\hp\lab3> notepad deployment.yaml
PS C:\Users\hp\lab3> kubectl apply -f deployment.yaml
deployment.apps/lab3-web configured
PS C:\Users\hp\lab3> kubectl get pods -o wide
NAME                        READY   STATUS        RESTARTS   AGE     IP          NODE             NOMINATED NODE   READINESS GATES
lab3-web-594df555dc-vr2w4   1/1     Running       0          13m     10.1.0.12   docker-desktop   <none>           <none>
lab3-web-594df555dc-w6mvf   1/1     Terminating   0          6m18s   10.1.0.13   docker-desktop   <none>           <none>
lab3-web-594df555dc-xtqdj   1/1     Running       0          13m     10.1.0.11   docker-desktop   <none>           <none>
lab3-web-68bddf4cb5-4bmd6   0/1     Running       0          4s      10.1.0.15   docker-desktop   <none>           <none>
lab3-web-68bddf4cb5-mzbdb   1/1     Running       0          11s     10.1.0.14   docker-desktop   <none>           <none>
PS C:\Users\hp\lab3>
```

PART E:

E1:Delete pod & recreate

```
                    Content-Type: text/html; charset=utf-8
                    Date: Sat, 25 Apr 2026 18:59:09 GMT
                    Server: Werkzeug/3.1.8 Python/3.12.13

                    Hello from Lab 3
Forms             : {}
Headers           : {[Connection, close], [Content-Length, 16], [Content-Type, text/html; charset=utf-8], [Date, Sat,
                    25 Apr 2026 18:59:09 GMT]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 16



PS C:\WINDOWS\system32> Invoke-RestMethod http://localhost:9090/health
ok
PS C:\WINDOWS\system32> kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
lab3-web-868cd7fbff-7qv2l   1/1     Running   0          8m49s
lab3-web-868cd7fbff-j4tlp   1/1     Running   0          8m49s
lab3-web-868cd7fbff-wns8v   1/1     Running   0          8m49s
PS C:\WINDOWS\system32> kubectl delete pod lab3-web-868cd7fbff-wns8v
pod "lab3-web-868cd7fbff-wns8v" deleted from default namespace
PS C:\WINDOWS\system32> kubectl get pods -w
NAME                        READY   STATUS    RESTARTS   AGE
lab3-web-868cd7fbff-7qv2l   1/1     Running   0          14m
lab3-web-868cd7fbff-f7hz8   1/1     Running   0          5m7s
lab3-web-868cd7fbff-j4tlp   1/1     Running   0          14m
```

Part E2:

```
Windows PowerShell
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PS C:\WINDOWS\system32> notepad deployment.yaml
PS C:\WINDOWS\system32> cd hp/lab3
cd : Cannot find path 'C:\WINDOWS\system32\hp\lab3' because it does not exist.
At line:1 char:1
+ cd hp/lab3
+ ~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (C:\WINDOWS\system32\hp\lab3:String) [Set-Location], ItemNotFoundExcepti
   on
    + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\WINDOWS\system32> cd C:\Users\hp\lab3
PS C:\Users\hp\lab3> notepad deployment.yaml
PS C:\Users\hp\lab3> kubectl apply -f deployment.yaml
error: error parsing deployment.yaml: error converting YAML to JSON: yaml: line 15: did not find expected key
PS C:\Users\hp\lab3> notepad deployment.yaml
PS C:\Users\hp\lab3> kubectl apply -f deployment.yaml
deployment.apps/lab3-web configured
PS C:\Users\hp\lab3> kubectl get pods
NAME                        READY   STATUS        RESTARTS   AGE
lab3-web-594df555dc-2pmf2   1/1     Running       0          20s
lab3-web-594df555dc-vr2w4   0/1     Running       0          2s
lab3-web-594df555dc-xtqdj   1/1     Running       0          11s
lab3-web-868cd7fbff-7qv2l   1/1     Running       0          36m
lab3-web-868cd7fbff-f7hz8   1/1     Terminating   0          27m
lab3-web-868cd7fbff-j4tlp   1/1     Terminating   0          36m
PS C:\Users\hp\lab3>
```

Concept Reflection:

1-Namespaces only isolate what a process can see ,but they do not limit how much CPU, memory, or I/O a process consumes.So,a container could still use excessive resources and affect other containers on the same host. Fairness requires resource control, not just isolation.

2-Cgroups enforce resource limits (CPU, memory, disk I/O) on containers. This prevents any single container from exhausting system resources.

3-Docker images are built in layers, which allows:

> **caching ,reuse ,smaller transfers**
>
> **In large-scale systems this improves deployment speed,storage efficiency,network usage**

4-In Kubernetes, desired state means the condition that I want my application to be in, like how many Pods should be running or which image to use. I define this in files like deployment.yaml. Kubernetes then keeps checking the system and makes sure the actual state always matches what I defined.

5-Self-healing in Kubernetes happens automatically, while in traditional systems I would have to fix problems manually.

6-Readiness and liveness probes have different purposes. Readiness checks if the application is ready to receive traffic, while liveness checks if it is still running properly.

7-In a single-node kind cluster, all the Pods run on the same node. In a real multi-node cluster, scheduling would involve choosing between different nodes.